VeriTwin

Don’t apply. Prove it.

A proof-of-work platform that turns real organizational tasks into 10-minute trials, and turns how a student thinks into portable, evidence-linked proof, so talent is found without a resume, a pedigree, or prior experience.

Challenge	AI Hackathon Briefing - 2 Hours	**Problem**	The Talent-Opportunity Gap
Document	Final idea brief (revised after critique)	**Build time**	2-hour demo, 3 screens

1. Problem Statement

Talented students stay invisible to the opportunities that would grow them, because the signals the market relies on (resumes, college names, prior experience) measure pedigree rather than capability. Three failures compound each other:

Failure	What happens	Who is hurt
Resumes are broken	AI-polished resumes all look identical, so they hide genuine problem-solvers who lack formal credentials or elite college tags.	First-gen and non-tier-1 students; recruiters who cannot tell candidates apart.
The experience catch-22	Opportunities ask for prior experience, but experience can only be gained from a first opportunity.	Every student without a first break; orgs that never see them.
Blind networking	Students cannot find peers, projects, and organizations that fit their working style and strengths, and teams cannot find complementary members.	Students, project teams, and small orgs with no recruiting budget.

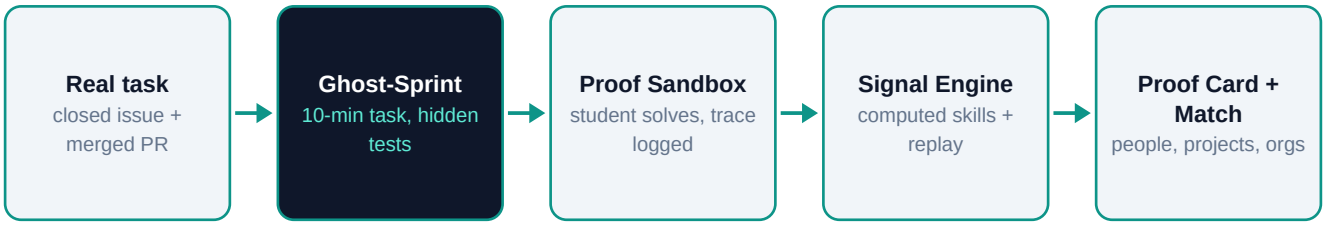
> **Problem statement in one line:** How might we let any student demonstrate real ability, and be discovered and matched on it, without needing a resume, a pedigree, or prior experience?

Why current platforms fail

- **Job boards and resume parsers** optimize keyword matching on the very signal that is broken.
- **Skill-test platforms** use synthetic puzzles disconnected from real work, and they only work in one direction: the employer screens, the student performs.
- **Social and portfolio networks** reward visibility and polish, not demonstrated reasoning.

2. The Solution on One Page

> **Thesis:** Real tasks from real organizations become verified proof-of-work for students. Organizations come to students with work, students answer with evidence, and matching follows the evidence.



Core loop: each step produces something the next step can trust. The task has a ground-truth answer, the trace is recorded by the platform itself, the skills are computed by code, and the match explains itself.

Three pillars

- **Ghost-Sprint (the core).** Organizations and open-source maintainers supply real, already-solved work. VeriTwin turns each item into a self-contained 5 to 10 minute trial with hidden tests.
- **Proof Sandbox + Signal Engine.** Students solve inside a browser sandbox. The platform records the work and computes skill signals deterministically. The result is a portable **Proof Card** the student owns.
- **Explainable matching.** Students are matched to teammates, projects, and organizations using evidence from their Proof Card, with a plain-language reason for every match.

What changed after the critique

Original draft	Final design	Why
Browser + IDE extension watching students passively	Recording only inside the VeriTwin sandbox; opt-in, student-owned, deletable	Removes the surveillance problem and the install friction
LLM reads live edits and plots a radar	Signals computed by code from logged events; LLM only narrates, and every claim links to a replay moment	Numbers are reproducible and explainable
Auto-sanitize private production bugs	Closed public issues with merged PRs (PR tests = ground truth); private-repo pipeline with human approval is roadmap	Ground truth and no leak risk in the demo
Anti-cheating framing	AI use is allowed and measured as a skill	Everyone uses AI; judgment with AI is the real signal
Agent-to-agent background simulations	Embeddings + constraints with explanations; Talent Agent and Scout Agent are UX personas	Buzzword replaced by something demonstrable
Five extra headline features	Two headline features (Thought-X-Ray, Learning Velocity); the rest moved to roadmap	Focus wins a 2-hour build
Fairness not addressed	Untimed mode, accommodations, no typing-speed signals, bias audit; scores are evidence, not verdicts	Avoids swapping resume bias for new bias
Long protocol name	Tagline: “Don’t apply. Prove it.”	Memorable in one breath

3. How It Works

3.1 Ghost-Sprint Generator

Step	What happens
1. Ingest	Take a closed issue and its linked merged PR from a public repo. Roadmap: private repos through an org-installed GitHub app.
2. Filter	Keep only tasks with a single PR, a small diff (about 3 files or fewer, under about 60 changed lines), test changes included, runtime under 30 seconds, and an issue that makes sense on its own.
3. Package	Starting state is the commit before the fix. Hidden tests come from the PR. The student sees the failing symptom, never the solution.
4. Rewrite	An LLM turns the issue into a self-contained 5 to 10 minute brief: context, expected behavior, difficulty estimate.
5. Verify	Automatically confirm the tests fail on the pre-fix commit and pass with the PR's patch. Tasks that fail verification are discarded.
6. Calibrate	Set the benchmark time from reference solutions and a pilot cohort, not from the maintainers' own time.

Private-repo sanitization (roadmap): *secret scanning, identifier renaming, LLM rewrite, and a mandatory human approval by the organization before anything is published.*

3.2 Proof Sandbox

A browser editor with a test runner. The platform logs timestamped events: edits (batched), test runs and results, reverts and rewrites, pastes, custom inputs the student tries, hint requests, and any AI prompts sent inside the sandbox. Nothing outside the sandbox is observed. The student owns the trace, can delete it at any time, and chooses what to share.

3.3 Signal Engine (computed by code, not guessed by an LLM)

Signal	Computed from logged events	What it indicates
Debugging speed	Time from first failing run to first fully passing run, normalized by the task's calibrated median	How quickly the student diagnoses and fixes
Edge-case awareness	Share of hidden edge-case tests passed, plus custom inputs the student ran on their own	Whether they probe boundaries, not just the happy path
Iteration discipline	Runs per meaningful edit, edit size between runs, revert and rewrite count	Systematic versus scattershot work
Pivot quality	Whether a pivot (large revert or rewrite after repeated failures) is followed by fewer failing tests	Grit combined with judgment
Learning velocity	Slope of time-to-green and hint use across a 3-task sequence that introduces one new concept	How fast they learn something unfamiliar
AI collaboration	Prompts sent, share of AI-pasted code later edited or tested, whether wrong AI output was caught	Judgment when working with AI

Typing speed is never a signal. All signals are indicative evidence, not a verdict on a person.

3.4 Thought-X-Ray: the 30-second replay

- Rules segment the trace into phases: read, hypothesize, edit, test, pivot.
- An LLM writes a short narration of the key moments. It may only cite logged event IDs; any sentence it cannot tie to an event is dropped.
- The recruiter sees where the student got stuck, what they tried next, and how the pivot worked, in about 30 seconds.

3.5 Learning-Velocity Radar (slope over status)

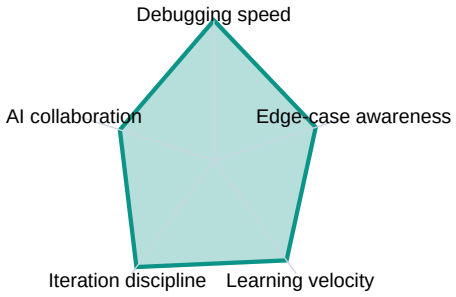
A short sequence of three tasks introduces one unfamiliar concept (for example, a small library the student has not used, with a doc snippet). The platform measures how time-to-green and hint use change across the sequence. It is shown as a trend with an honest confidence band, and needs at least three data points before it is displayed as a score.

3.6 Matching Engine

- **Inputs:** the student's skill vector from the Proof Card; project or organization requirement vectors; constraints such as availability and domain.
- **Method:** embedding similarity plus constraint filters for people-to-project matches; for teams, maximize coverage of required skills and diversity of working style.
- **Explainability:** every match carries a reason tied to evidence, for example "matched because you fixed an async race condition in 6 minutes with 2 test runs."
- **Personas:** the Talent Agent (speaks for the student) and the Scout Agent (speaks for the project) are the user-facing experience of this engine.

3.7 Fast-Track Unlock

An organization sets a threshold and optionally a sponsor-funded micro-stipend. Beating the benchmark queues an invitation for **human review**. It is a fast path to a conversation, not an automatic hire.



Sample Proof Card (illustrative)

- **Debugging speed:** first failing run at 02:10, all tests green at 06:40, against a calibrated median of 09:00. [replay 02:10 to 06:40]
- **Pivot:** reverted a caching fix after 3 failed runs and found the real cause in the input parser. [replay 04:05]
- **AI collaboration:** 2 prompts; edited the suggested patch and re-ran tests before accepting.
- **Learning velocity:** time-to-green fell across the 3-task sequence (11:20, 7:05, 4:30).

All numbers above are illustrative sample data for layout only. In the product, every claim links to a real moment in the student's replay.

4. Feature Scope

Tier	Feature	Notes
Headline	Thought-X-Ray 30-second replay	The most demoable moment; evidence-linked narration
Headline	Learning-Velocity Radar	Slope over status; shown only with enough data points
Supporting	Proof Card	Portable, student-owned, shareable by link
Supporting	Explainable match	Teammates, projects, organizations, each with a reason
Supporting	AI-collaboration signal	Measures judgment with AI instead of banning it
Roadmap	Complementary Squad Synthesizer	Graph-based team formation from skill coverage and working style
Roadmap	Shadow Pitch and non-code Ghost-Sprints	Rubric-graded case tasks (analysis, design teardown) plus an AI challenger for design, business, and research students; v1 is coding-first
Roadmap	Private-repo Ghost-Sprints	Secret scanning, renaming, LLM rewrite, human approval
Roadmap	Funded Bounty Unlock	Sponsor-funded micro-stipends with human review

5. Responsible Design

Privacy and consent

- No extension, no passive tracking. Recording happens only inside the sandbox, with explicit opt-in.
- Students own their traces: view, export, delete, and choose who sees their Proof Card.
- Designed around consent and data-minimization principles (for example, India's DPDP Act). A legal review is needed before any real launch.

Fairness

- Untimed mode and accommodation options; no typing-speed or hardware-dependent signals.
- Signals are normalized per task and shown as evidence, not as a single ranking of people by default.
- Regular bias audits across cohorts (college tier, language background, device and connectivity).

Anti-gaming and integrity

- Multiple independent signals and a replay make a single trick hard to hide; tasks rotate and can be re-generated as variants.
- AI use is allowed and measured; a short live follow-up at the fast-track step confirms understanding.
- The LLM never produces numbers and may only cite logged events, which limits hallucinated claims.

6. How VeriTwin Differs

Dimension	Resume + job boards	Coding-test platforms	Portfolios (GitHub, Devpost)	VeriTwin
Task source	None	Synthetic puzzles	Self-chosen projects	**Real organizational work**
What is measured	Keywords, pedigree	Correct answer, speed	Output of past work	**How the student reasons and learns**
Direction	Student applies	Employer screens	Student publishes, waits	**Organizations come to students**
Evidence for recruiter	Self-reported	Score	Repo, hard to compare	**Replay, linked to moments**
Helps students with no experience	No	Partly	Only if they already built things	**Yes: the first trial creates the first proof**

Comparison reflects the general category of each approach, not any specific vendor's full feature set.

7. The 2-Hour Build and Demo Plan

Suggested stack

- **Frontend:** React or plain HTML with the Monaco editor; **runtime:** Pyodide so Python tests run in the browser.
- **Backend:** a small FastAPI or serverless service; GitHub REST API for issues, PRs, and test files; any LLM API for task rewrite and narration.
- **Storage and matching:** JSON or SQLite; sentence embeddings with cosine similarity and simple constraint filters.
- **Safety net:** cache the chosen issues and 3 pre-seeded student traces so the demo never depends on live network calls.

Timeline

Time	Work	Output
0:00 to 0:15	Pick 2 closed issues with merged PRs from small Python repos; cache JSON; app skeleton	Task data + skeleton
0:15 to 0:55	Monaco sandbox, Pyodide test runner, event logger (runs, edits, reverts, pastes, timestamps)	Working Proof Sandbox
0:55 to 1:20	Ghost-Sprint generator: fetch issue and PR, LLM brief, starter state, hidden tests, fail-to-pass check	Screen 2
1:20 to 1:45	Signal engine, replay timeline, narration with event-ID check, radar from computed features	Screen 1 + Proof Card
1:45 to 2:00	Match screen on seeded profiles; rehearse; record a fallback video	Screen 3 + backup

Cut line if behind schedule: replace the live runtime with pre-recorded traces, but keep the signal computation and the replay real.

The three demo screens

Screen	What the judge sees	Real vs pre-seeded
1. Student Workspace	Editor with failing tests; a side panel counting runs and time-to-green; on completion, the radar and Proof Card appear with replay links.	Live run; radar computed from the real trace
2. Ghost-Sprint Generator	Paste a closed issue or PR link; get a 10-minute brief, starter state, hidden-test summary, difficulty and time estimate, and an org-approval toggle.	Live fetch with cached fallback
3. Match Reveal	From the Proof Card: 3 complementary teammates and 2 projects or organizations, each with a “why matched” reason; a fast-track banner when the benchmark is beaten, marked as pending human review.	Seeded profiles, real matching logic

8. Pitch

90-second script

> *Two students apply for the same internship. Same resume, polished by the same AI. One went to a famous college. Who do you pick? Today, the answer is the college. That is the Talent-Opportunity Gap.*
>
> *VeriTwin changes the question from "where did you study?" to "show me how you think." Organizations and open-source maintainers publish real, already-solved issues. We turn each into a 10-minute Ghost-Sprint with hidden tests. A student solves one in our sandbox, and instead of a resume we get a Proof Card: a 30-second replay of where they got stuck, how they pivoted, and how fast they learn, with every claim linked to the exact moment.*
>
> *Then we match on that evidence: teammates who complement, projects and organizations that need exactly this. Beat the benchmark, and a human reviewer fast-tracks you. No resume. No pedigree. No catch-22.* ***Don't apply. Prove it.***

Questions judges are likely to ask

Question	Answer
Isn't this HackerRank with a replay?	The tasks are real organizational work, not puzzles; the direction is reversed (organizations publish, students respond); and matching runs on linked evidence rather than a score.
How do you stop cheating or AI use?	We allow AI and measure how it is used. Tasks rotate, multiple signals cross-check each other, and a short live follow-up happens at the fast-track step.
Is the radar real or LLM-made?	Computed by code from logged events, with each axis defined. The LLM only narrates and may cite only logged events. Scores are evidence, not verdicts.
What about privacy?	No extension. Recording only in our sandbox, opt-in, student-owned, deletable, shared only by the student's choice.
Why would organizations join?	Open-source maintainers get vetted contributors and triaged issues, and need no integration. Startups get pre-verified shortlists without running their own screening.
Is it fair to all students?	Untimed mode, accommodations, no typing-speed signals, per-task normalization, and bias audits across cohorts.
What about non-coders?	v1 is coding-first. Roadmap adds rubric-graded case Ghost-Sprints and Shadow Pitch for design, business, and research students.
How do you know it predicts performance?	We do not claim it yet. A pilot with a placement cell will correlate signals with later internship and project outcomes and report the results honestly.

9. Go-to-Market, Business Model, and Metrics

Cold-start strategy

- **Phase 1 (first months):** open-source maintainers (public issues, no legal friction, real incentive to find contributors) and one or two college placement cells or coding clubs.
- **Phase 2:** startups and labs through private Ghost-Sprints with the sanitization pipeline and human approval.
- **Phase 3:** non-code tracks, Squad Synthesizer, and sponsor-funded stipends.

Business model

- Students: free. Organizations: free for public and open-source tasks; paid for private Ghost-Sprints, shortlist delivery, and ATS integration.
- Institutions: dashboards for placement cells that show cohort proof and gaps.

Success metrics

- Time from sign-up to first real opportunity; share of fast-track invitations going to students outside top-tier colleges.
- Ghost-Sprint completion rate; organization time saved per hire.
- Predictive validity: correlation between signals and later internship or project outcomes in the pilot.

10. Risks and Mitigations

Risk	Mitigation
Signals do not predict real performance	Position signals as evidence; run a validation pilot; publish results; retire signals that do not hold up.
New bias replaces resume bias	Untimed mode, accommodations, no typing-speed signals, cohort bias audits, evidence-first display.
Two-sided cold start	Start with public open-source issues (no integration needed) and placement cells.
Brittle or leaky tasks	Strict task filters; automatic fail-to-pass verification; private-repo tasks only with scanning and human approval.
LLM hallucination in narration	LLM produces no numbers and may only cite logged event IDs; unsupported sentences are dropped.
Privacy backlash	In-sandbox recording only, opt-in, student-owned data, deletion on request, legal review before launch.
Live demo failure	Cached issues, pre-seeded traces, a recorded fallback video, and a clear cut line in the plan.

> **One-line pitch:** VeriTwin turns real organizational tasks into 10-minute trials, so any student can prove how they think, and get found for it, without a resume. **Don’t apply. Prove it.**